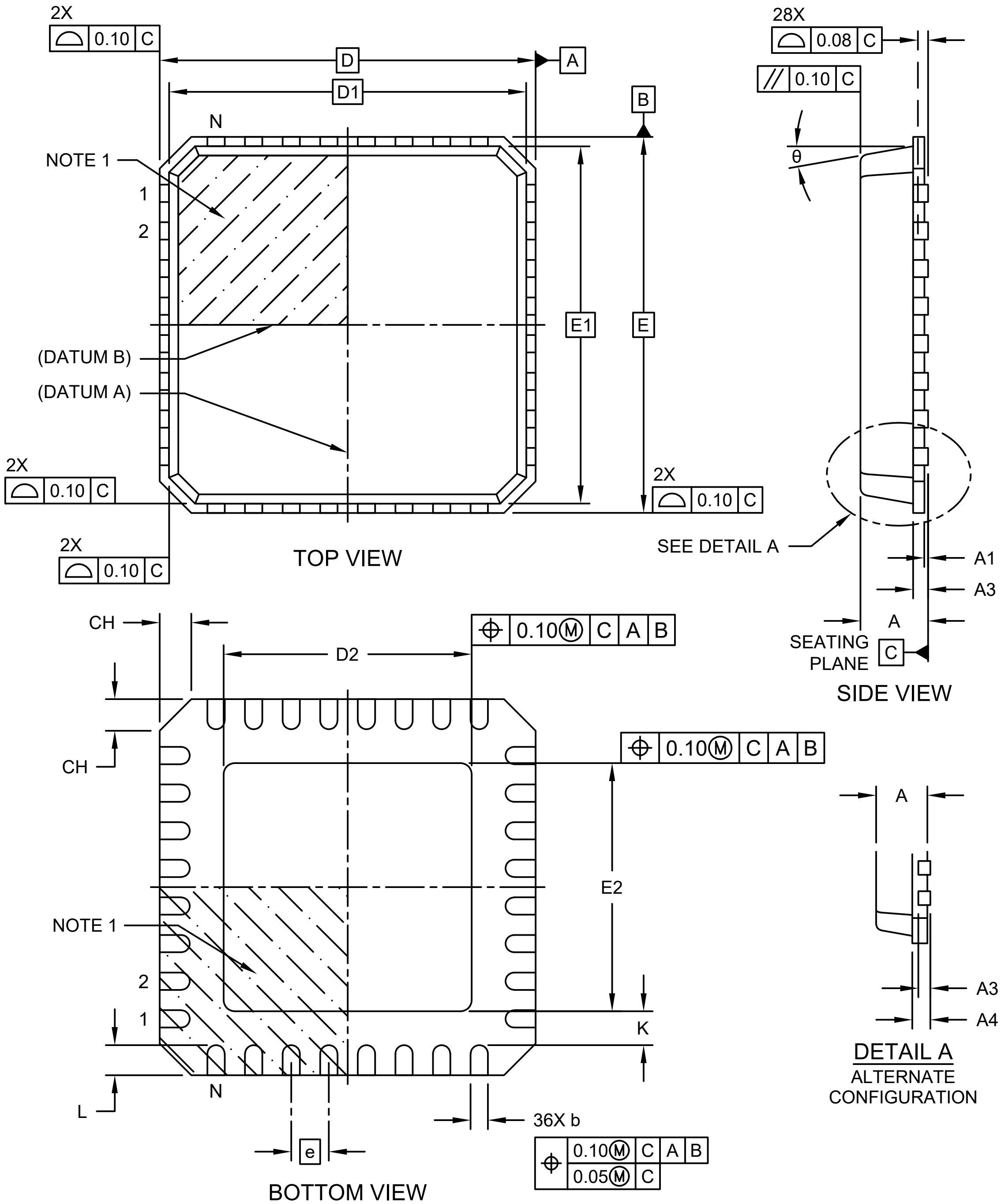
32-Lead Very Thin Plastic Quad Flat, No Lead Package (RN) - 5x5 mm Body [VQFN] With 3.3x3.3 mm Exposed Pad, Punch Singulated; Formerly called QFN

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



Microchip Technology Drawing C04-232-RN Rev B Sheet 1 of 2

32-Lead Very Thin Plastic Quad Flat, No Lead Package (RN) - 5x5 mm Body [VQFN] With 3.3x3.3 mm Exposed Pad, Punch Singulated; Formerly called QFN

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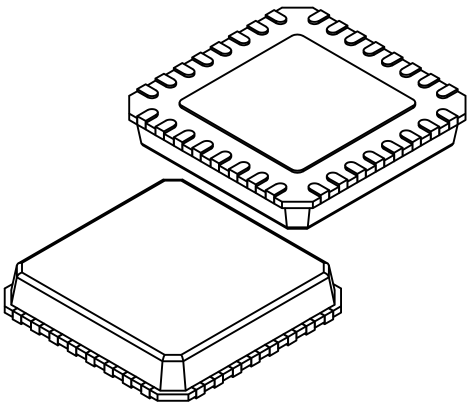
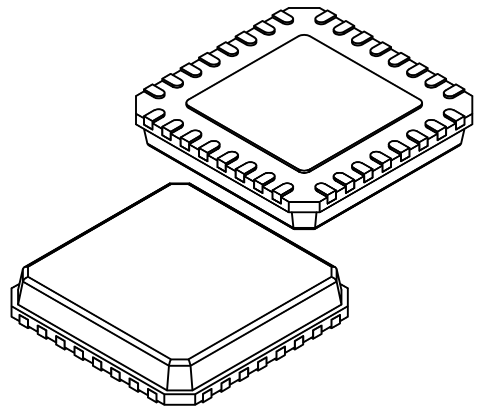


Original Configuration



Alternate Configuration

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N		32	
Pitch	e		0.50 BSC	
Overall Height	A	0.80	0.85	0.90
Standoff	A1	0.00	0.02	0.05
Terminal Thickness	A3		0.20 REF	
Overall Length	D		5.00 BSC	
Mold Cap Length	D1		4.75 BSC	
Exposed Pad Length	D2	3.20	3.30	3.40
Overall Width	E		5.00 BSC	
Mold Cap Width	E1		4.75 BSC	
Exposed Pad Width	E2	3.20	3.30	3.40
Chamfer	CH	–	–	0.60
Terminal Width	b	0.18	0.23	0.30
Terminal Length	L	0.30	0.40	0.50
Terminal-to-Exposed-Pad	K	0.20	-	-
Mold Draft Angle	θ	–	–	14°

ALTERNATE CONFIGURATION

Overall Height	A	0.80	0.85	0.90
Terminal Thickness	A3		0.20 REF	
Base Thickness	A4		0.30 REF	
Mold Cap Length	D1		4.71 BSC	
Mold Cap Width	E1		4.71 BSC	

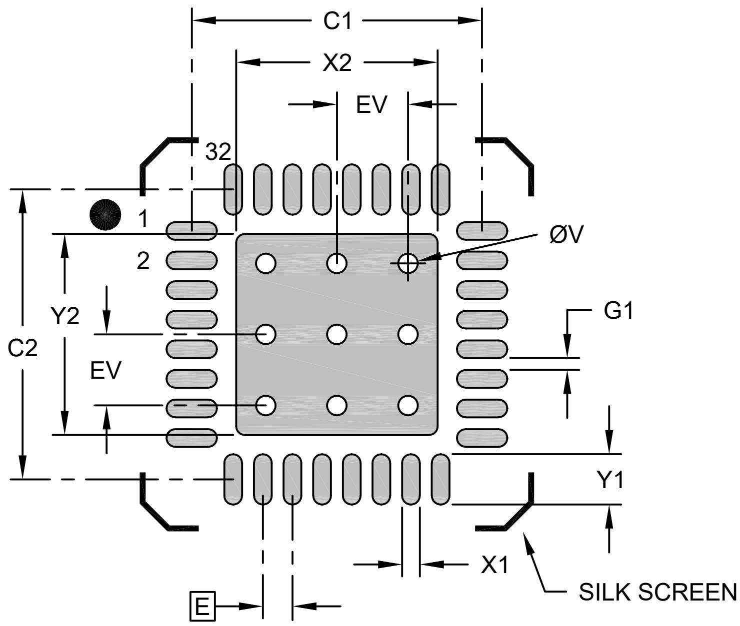
Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Package is punch singulated
3. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
   REF: Reference Dimension, usually without tolerance, for information purposes only.

Microchip Technology Drawing C04-21409-RN Rev B Sheet 2 of 2

32-Lead Very Thin Plastic Quad Flat, No Lead Package (RN) - 5x5 mm Body [VQFN] With 3.3x3.3 mm Exposed Pad, Punch Singulated; Formerly called QFN

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E		0.50 BSC	
Optional Center Pad Width	X2			3.40
Optional Center Pad Length	Y2			3.40
Contact Pad Spacing	C1		4.90	
Contact Pad Spacing	C2		4.90	
Contact Pad Width (X32)	X1			0.30
Contact Pad Length (X32)	Y1			0.85
Space Between Contacts (X28)	G1	0.20		
Thermal Via Diameter	V		0.33	
Thermal Via Pitch	EV		1.20	

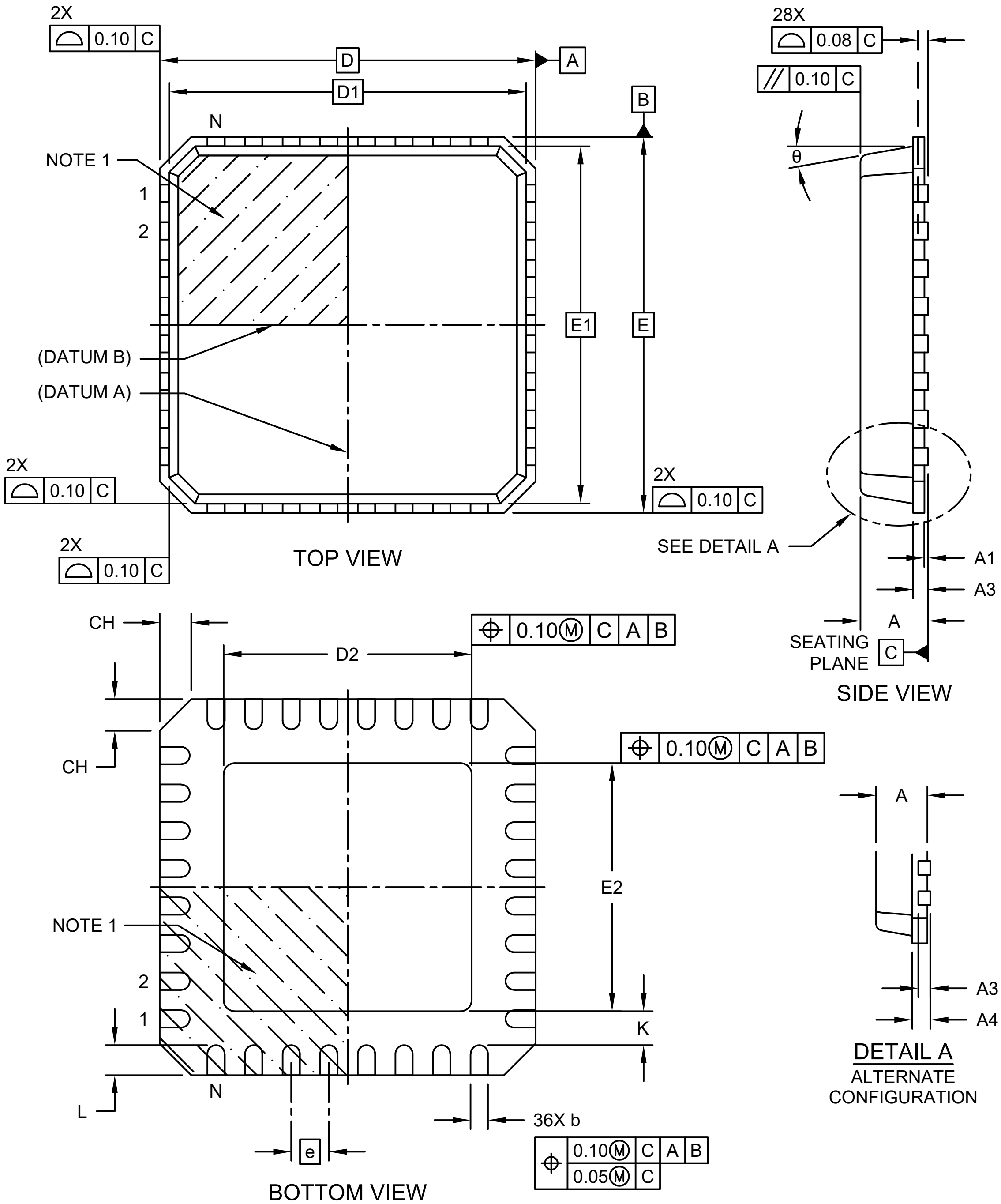
Notes:

1. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
2. For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-2232-RN Rev B

32-Lead Very Thin Plastic Quad Flat, No Lead Package (RNX) - 5x5 mm Body [VQFN] With 3.3x3.3 mm Exposed Pad, Punch Singulated; Formerly called QFN

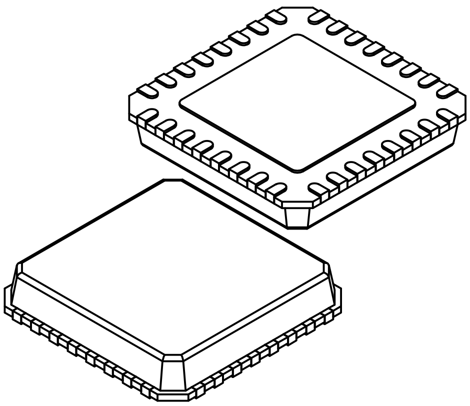
Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



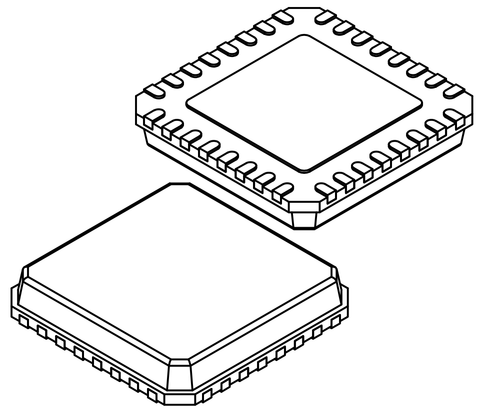
Microchip Technology Drawing C04-232-RNX Rev B Sheet 1 of 2

32-Lead Very Thin Plastic Quad Flat, No Lead Package (RNX) - 5x5 mm Body [VQFN] With 3.3x3.3 mm Exposed Pad, Punch Singulated; Formerly called QFN

Note: For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



Original Configuration



Alternate Configuration

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N		32	
Pitch	e		0.50 BSC	
Overall Height	A	0.80	0.85	0.90
Standoff	A1	0.00	0.02	0.05
Terminal Thickness	A3		0.20 REF	
Overall Length	D		5.00 BSC	
Mold Cap Length	D1		4.75 BSC	
Exposed Pad Length	D2	3.20	3.30	3.40
Overall Width	E		5.00 BSC	
Mold Cap Width	E1		4.75 BSC	
Exposed Pad Width	E2	3.20	3.30	3.40
Chamfer	CH	–	–	0.60
Terminal Width	b	0.18	0.23	0.30
Terminal Length	L	0.30	0.40	0.50
Terminal-to-Exposed-Pad	K	0.20	-	-
Mold Draft Angle	θ	–	–	14°

ALTERNATE CONFIGURATION

Overall Height	A	0.80	0.85	0.90
Terminal Thickness	A3		0.20 REF	
Base Thickness	A4		0.30 REF	
Mold Cap Length	D1		4.71 BSC	
Mold Cap Width	E1		4.71 BSC	

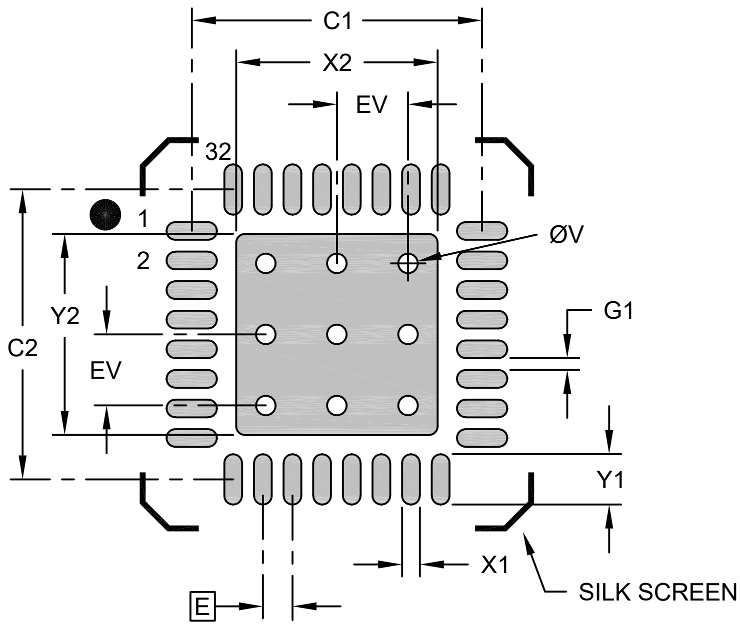
Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Package is punch singulated
3. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
   REF: Reference Dimension, usually without tolerance, for information purposes only.

Microchip Technology Drawing C04-21409-RNX Rev B Sheet 2 of 2

32-Lead Very Thin Plastic Quad Flat, No Lead Package (RNX) - 5x5 mm Body [VQFN] With 3.3x3.3 mm Exposed Pad, Punch Singulated; Formerly called QFN

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



RECOMMENDED LAND PATTERN

	Units	MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E		0.50 BSC	
Optional Center Pad Width	X2			3.40
Optional Center Pad Length	Y2			3.40
Contact Pad Spacing	C1		4.90	
Contact Pad Spacing	C2		4.90	
Contact Pad Width (X32)	X1			0.30
Contact Pad Length (X32)	Y1			0.85
Space Between Contacts (X28)	G1	0.20		
Thermal Via Diameter	V		0.33	
Thermal Via Pitch	EV		1.20	

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
2. For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-2232-RNX Rev B